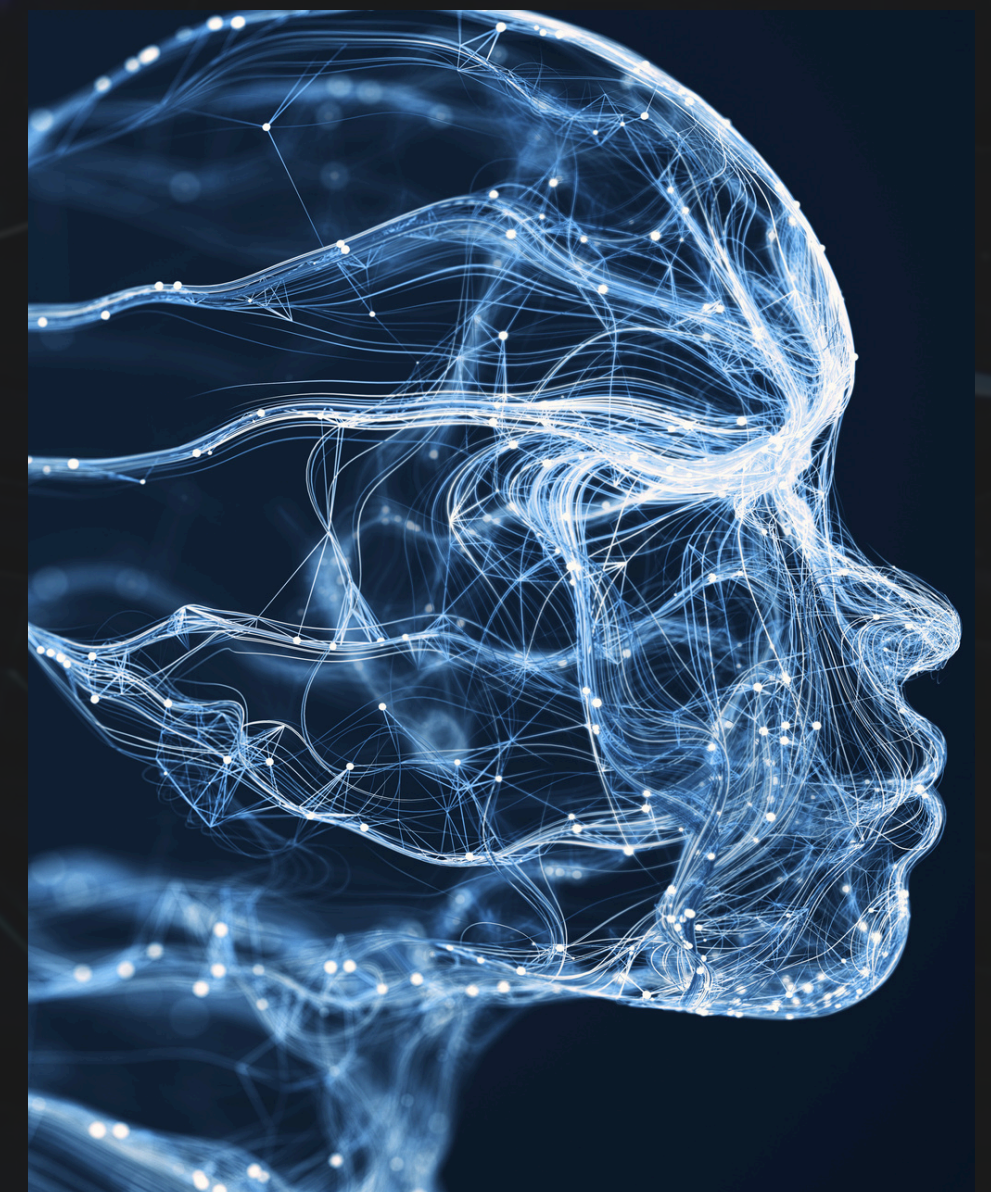




PL NEURAL NETWORK FINAL

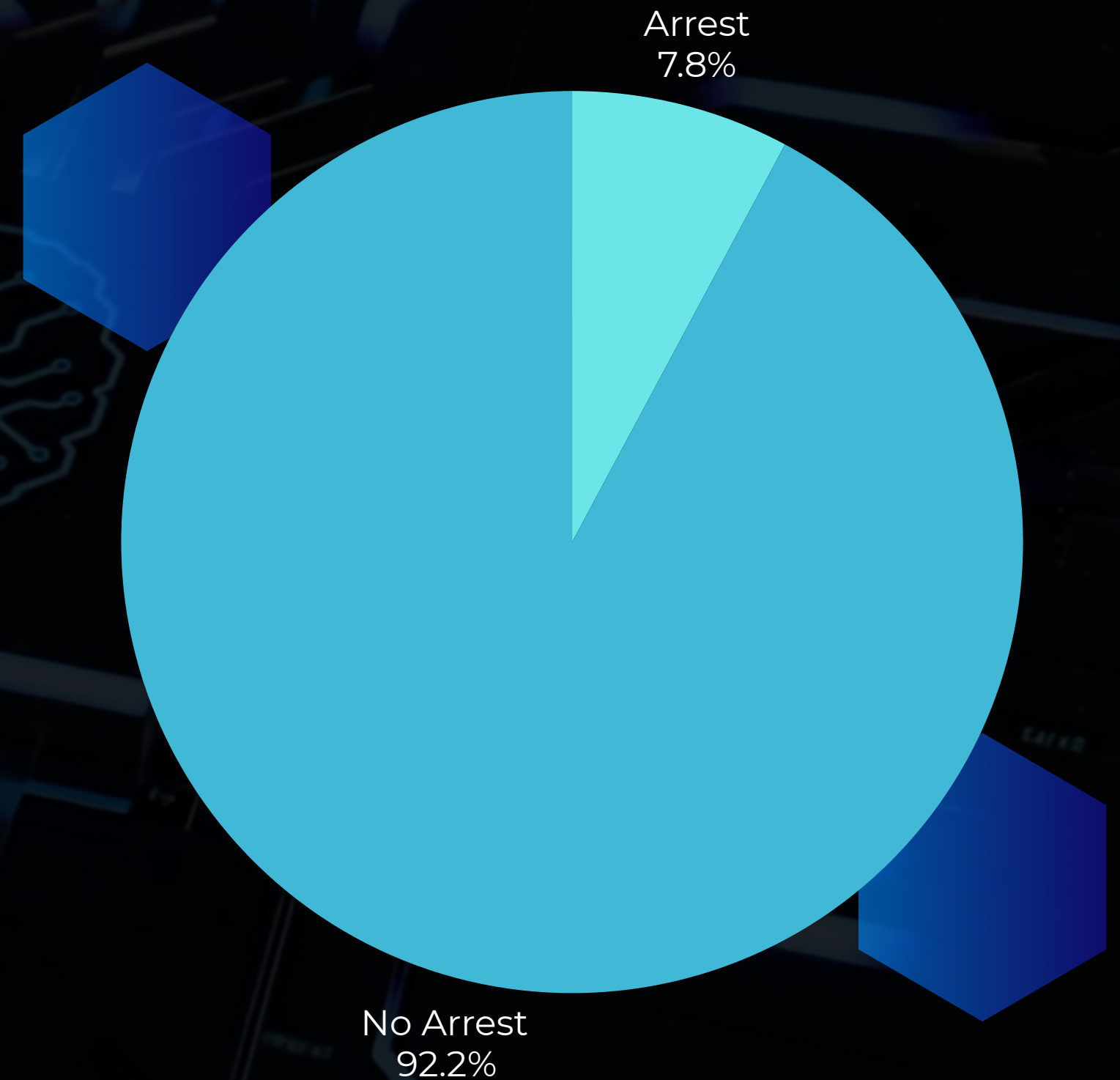
Created by: Noah Gallego, Braden Smelser, Marlon Elvir,
Jonathan Torres





DATA SNAPSHOT

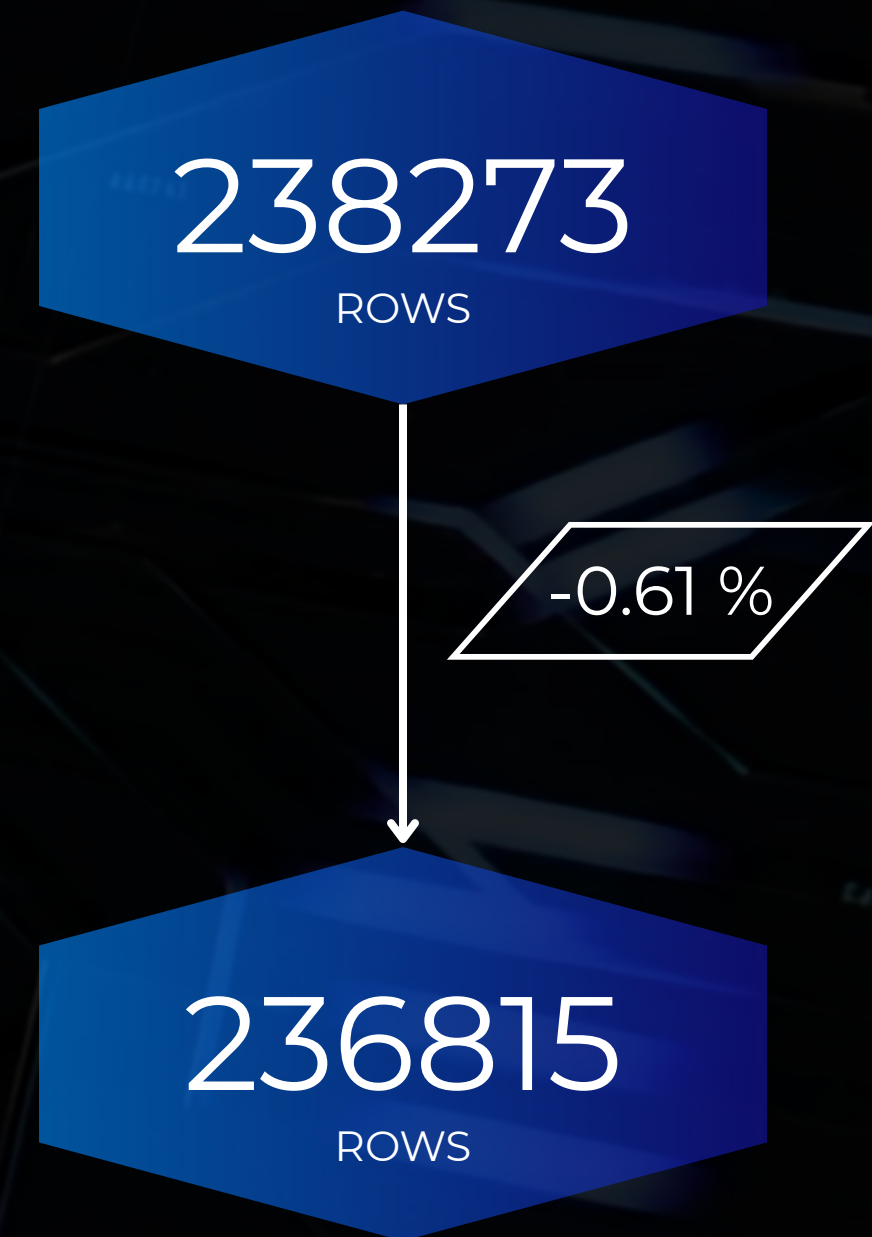
- Data collected from 2023 to Present.
- Data was subdivided in a 80/20 split.
- The target column was heavily imbalanced as shown to the right.
- After data cleaning, roughly 300k total rows.
- Highly Categorical Data.
- 21 total features provided.





DATA CLEANING: OUTLIER REMOVAL

- We noticed that there were few outliers, making up only 0.61% of the data in total.
- We iterated through each numerical column, and removed the outlier using the IQR (Inter-Quartile Range) method.





DATA CLEANING: NA / MISSING VALUES

- There were a few issues with our data such as Missing Values, Duplicate Rows, and Not Applicable (NA) values. To combat this we mainly dropped rows due to the percentage of data it took up.
 - Dropped extra index column, set DR_NO as index → slimmer file
 - Parsed date strings & fixed TIME_OCC ("0816" → "08:16")
 - Imputed 71% missing weapon fields with 0 / "No weapon identified"
 - Removed "X/H" gender, "-" descent, and any null victim info
 - Deleted 31k duplicate rows (-8%)
 - Result: 236k rows · 29 clean features · 0% NA



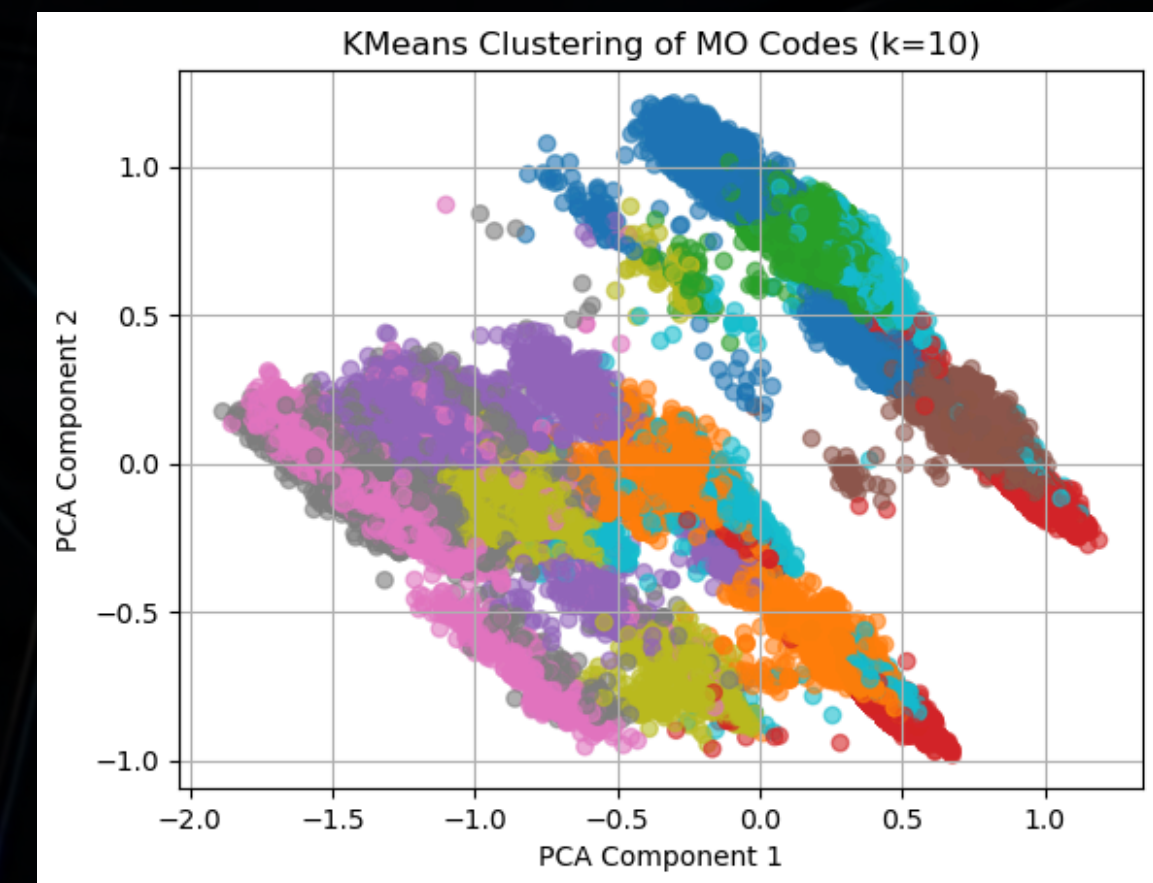
DATA CLEANING: FEATURE ENGINEERING (PT 1)

- To give the model more capacity to learn, we created the following features:
 - **Date & time decomposition:** Y/M/D/H/min + 4-bucket interval + 6-class time-of-day
 - **Seasonality:** days_to_holiday (distance to nearest US holiday)
 - **Demographics:** cleaned age, sex (M/F), descent (19 classes)
 - **Context:** area code/name, reporting district, premise & weapon codes/descriptions
 - **One-hot encoded Categoricals** → ~560 sparse columns
 - **Binary target:** Arrest / No-Arrest



DATA CLEANING: FEATURE ENGINEERING (PT 2)

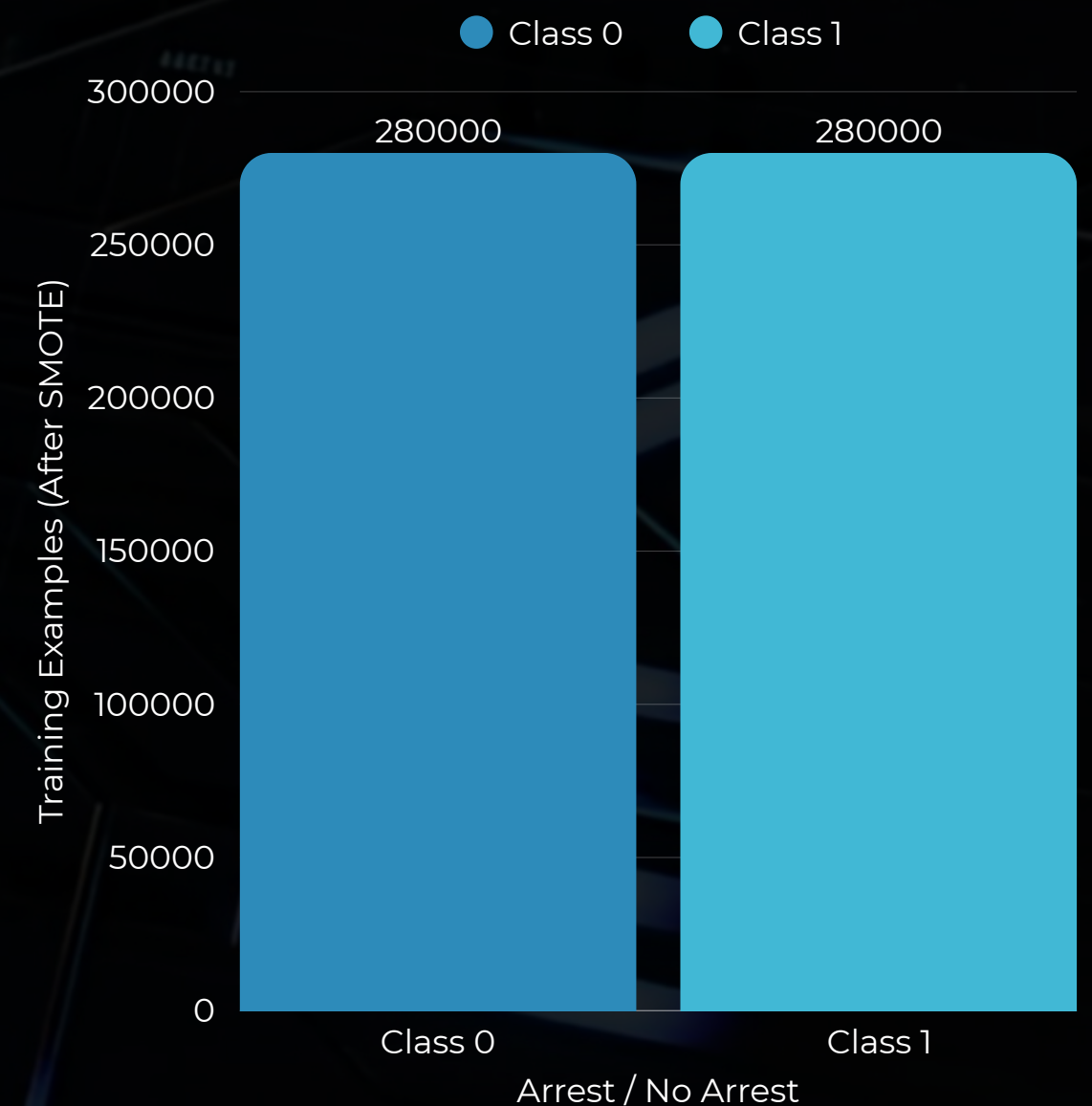
- While most columns were easy to One-hot encode, Mo-Codes posed a new challenge: 94k unique values. We extracted value out of mo_codes by:
 - Creating mo_code_count = #codes per incident
 - Top-20 codes → 20 binary flags (mo_1822, mo_0344, ...) (How many flags are present)
 - TF-vectorize + K-means (k=10) → mo_cluster behavior group
 - Net new MO features: 22 (count + flags + cluster)
 - Dropped raw text column after extraction





HANDLING CLASS IMBALANCE

- As shown in a previous slide, our target class has an extremely uneven distribution. To combat this we:
 - **Performed SMOTE (Synthetic Minority Oversampling Technique):** To synthetically generate rows in the minority class for the training set only.
 - **Used Focal Loss:** To generate a stronger penalty for inaccurate minority class predictions.
 - Tuned using α (alpha) = 0.8 & γ (gamma) = 1.0





MODEL ARCHITECTURE



Transformer Model

- Tab (Tabular) Transformer
- Same architecture that powers ChatGPT ('T' Stands for Transformer)
- Attention: Allows the model to receive context from every other feature in each row.
- Conditional Feature importance (weights less important features proportionally less)
 - 2 Encoder Layers
 - 4 Heads
 - 128 Hidden Units



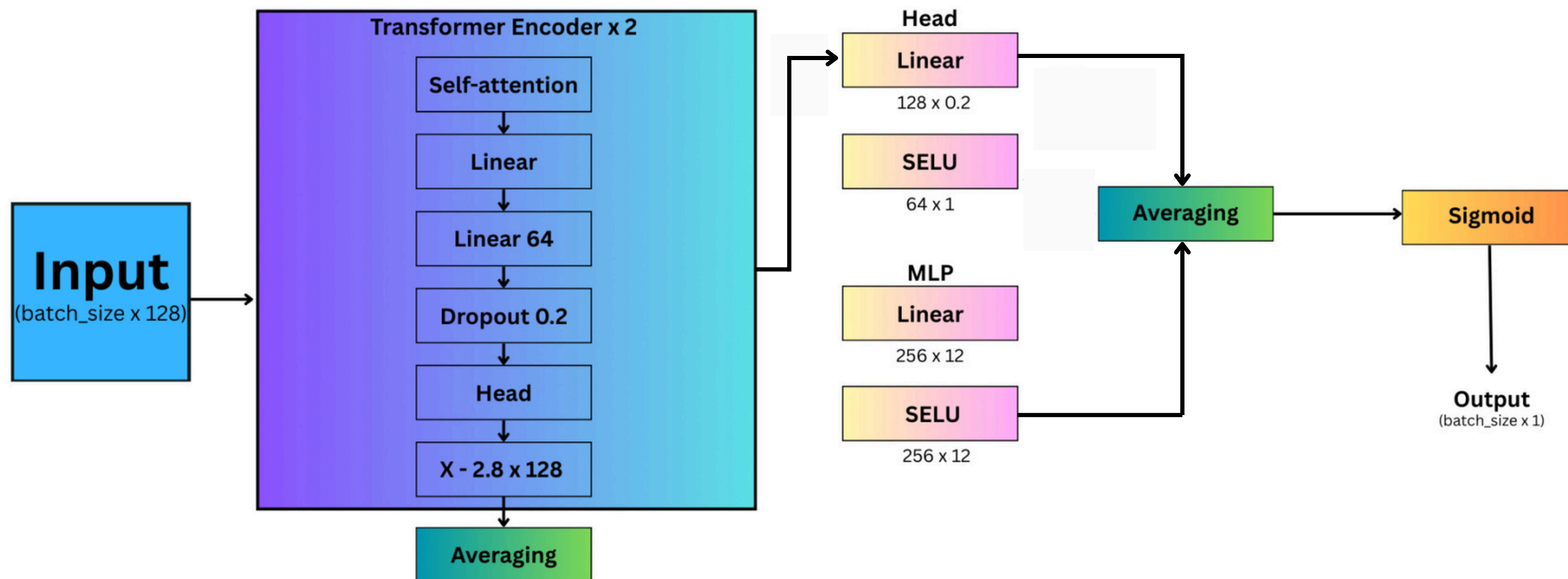
Multi-Layer Perceptron (MLP)

- Allows Model to capture features that the Transformer could not.
- 3-Layer architecture.
- SELU (Self-Normalizing Activation Units).
- 20% Dropout to prevent any overfitting.



Ensemble Learning

- Took two weaker models, and averaged the outputs for a more powerful prediction.
- Denoted as $\text{Sigmoid}((\text{mod1} + \text{mod2}) / 2)$





RESULTS

Predictions

- Accuracy: 0.903
- Precision: 0.963
- Recall: 0.933
- F1Score: 0.948
- ROCAUC: 0.835

No Arrest

Arrest

No Arrest

42046

1111

Arrest

1606

3042



THANK YOU!
DEMO TIME!